

adjust the Biped such that it adapts to the new position. These parameters are adjusted in the Key Info rollout, under Body. Balance Factor controls where along the spine the Biped's weight is cantered. Dynamics Blend controls how gravity affects the Biped when airborne. Ballistic Tension controls how the knee bends to absorb a jump or run step.

The Dynamics and Adaptation rollout allows you to use Biped's dynamics engine and set the value of Gravity. Spline Dynamics turns off the dynamics engine and uses spline based animation to calculate keyframes.

8.3.4 Animation Workbench

The Animation Workbench is a custom window for manipulating Biped animation curves. It works in the same way as Track view, and has the same Curve Editor, but it also has additional features for working with Biped curves. These tools include analysis functions to check curves for errors and automatically fix them. This is particularly helpful when working with motion capture data, or any animation that is key frame intensive.

8.4 Footstep Animation

Footsteps enable you to take advantage of Biped's built-in dynamics to create simulated human motion. The basic procedure is to lay down footsteps within the scene and then activate them. After this happens, the Biped will automatically walk from footstep to footstep. The walks, runs, and jumps initially created by using footsteps can be somewhat generic, but subsequent key frame animations can add characters to the original animation. Because footsteps are dynamically driven, key frame animation on certain parts of the character is restricted.

There are two methods of creating footsteps (single or multiple) along with three types of footsteps—walk, run, and jump. The different types of footsteps represent different timings for the footsteps.

- **Footstep mode**

Footstep mode can be entered by using either the Sub-Object pull-down or the Footstep icon.

- **Create Footsteps (Append)**

Allows you to lay down footsteps interactively. This appends the new footsteps after the last footstep in the animation.

- **Create Footsteps (At Current Frame)**

The same as Append, but creates the new footsteps at the current frame. If the new footstep overlaps with an existing footstep, an alert appears.